Recursion and stack

a1

a2

```
#include <stdio.h>

unsigned long long fact(unsigned x) {
    if (x == 0)
        return 1;

    return x * fact(x-1);
}

int main(void) {
    printf("!3 =: %llu\n", fact(3) );
}
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```

main	
return address: ?	return value: ?

Recursion and stack

fact	
x : 3	
return address: ?	return value: ?
main	
return address: a2	return value: ?

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Recursion and stack

fact	
x : 2	
return address: ?	return value: ?
fact	
x : 3	
return address: a1	return value: ?
main	
return address: a2	return value: ?

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    #include <stdio.h>

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```

Recursion and stack

fact	
x : 1	
return address: ?	return value: ?
fact	
x : 2	
return address: a1	return value: ?
fact	
x : 3	
return address: a1	return value: ?
main	
return address: a2	return value: ?

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Recursion and stack

<table>
<tr><td colspan="2">fact</td></tr>
<tr><td colspan="2">x : 0</td></tr>
<tr><td>return address: ?</td><td>return value: ?</td></tr>
<tr><td colspan="2">fact</td></tr>
<tr><td colspan="2">x : 1</td></tr>
<tr><td>return address: a1</td><td>return value: ?</td></tr>
<tr><td colspan="2">fact</td></tr>
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<tr><td>return address: a1</td><td>return value: ?</td></tr>
<tr><td colspan="2">fact</td></tr>
<tr><td colspan="2">x : 3</td></tr>
<tr><td>return address: a1</td><td>return value: ?</td></tr>
<tr><td colspan="2">main</td></tr>
<tr><td>return address: a2</td><td>return value: ?</td></tr>
</table>

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Recursion and stack

fact	
x : 1	
return address: a1	return value: 1
fact	
x : 2	
return address: a1	return value: ?
fact	
x : 3	
return address: a1	return value: ?
main	
return address: a2	return value: ?

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Recursion and stack

fact	
x : 2	
return address: a1	return value: 1
fact	
x : 3	
return address: a1	return value: ?
main	
return address: a2	return value: ?

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Recursion and stack

fact	
x : 3	
return address: a1	return value: 2
main	
return address: a2	return value: ?

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main	
return address: a2	return value: 6

Recursion and stack

printf	
d : 6	
return address: ?	return value: ?
main	
return address: a3	return value: ?

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main	
return address: a3	return value: 8

Recursion and stack

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